

01 / THE RESEARCH ANSWER

Cushing Inventory Stress

How does seasonal inventory relate to the WTI curve, and where does the relationship fail?

PUBLIC MARKET-STATE EVIDENCE

Seasonal stocks describe curve states; nonlinear evidence is weak after year controls.

Across 481 publication observations, the median F2-F3 quote is +0.37 USD/bbl in low-inventory states and -0.62 in high-inventory states (rank correlation -0.73). This historical association varies across years. The exploratory joint test does not provide sufficient evidence of a shape difference beyond the linear controlled relationship. Actual monthly contract data is still required to test incremental forecasting value and trading costs.

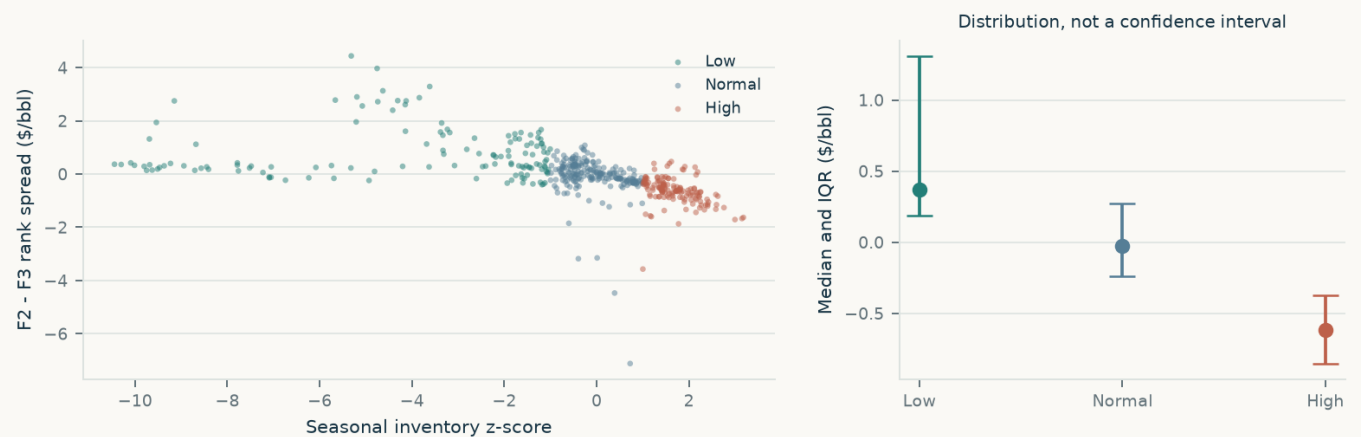


Figure 1. Newly reported seasonal inventory states and the preceding public F2-F3 quote. State medians summarize differences across the historical observations. Whiskers are distributions, not confidence intervals.

MATCHED RELEASES

481

POOLED SPEARMAN RHO

-0.729

2019: WEAK-YEAR RHO

0.079

Public price overlap: 07 Jan 2015 to 03 Apr 2024. Inventory records continue through release 10 Sep 2026.

Commercial use: treat inventory as a prompt for checking deliverable barrels and expected flows. A stock level does not establish that a spread will keep strengthening.
Predictive and trading evidence is pending because verified actual-contract market inputs are not yet available. No conclusion about forecast skill follows from this missing test.

02 / WHAT DOES INVENTORY STRESS MEAN?

A statistical state is not a tank limit.

$z = (\text{reported inventory} - \text{historical seasonal mean}) / \text{historical seasonal standard deviation}$. The reference uses the previous three calendar years, within 28 days either side of the same point in the year, and at least 20 observations known by the release cutoff. Low: $z < -1$; High: $z > 1$; Normal: -1 to $+1$. Low means below the historical seasonal reference, not a physical operating minimum or a capacity utilization measure.

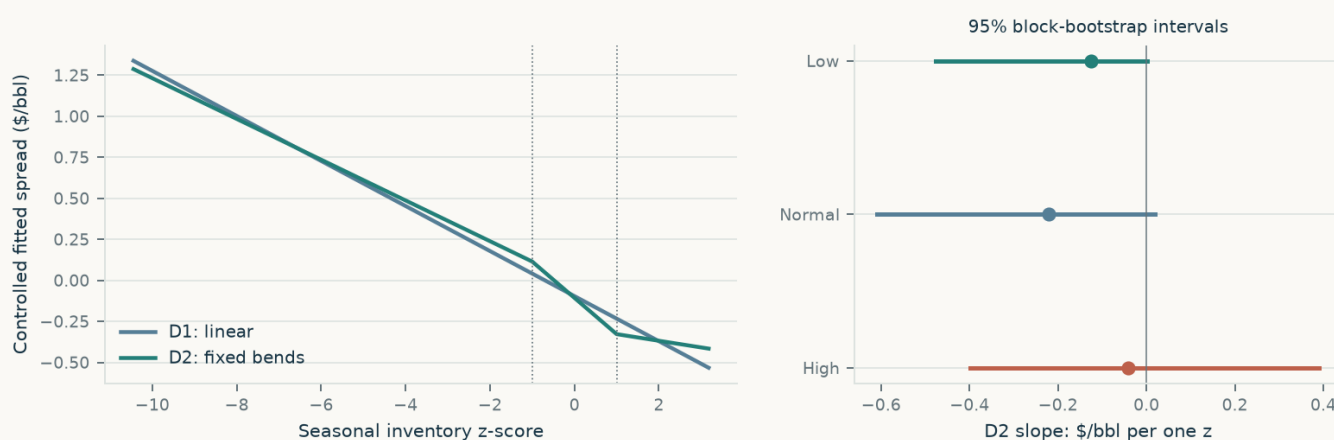


Figure 2. D1 is linear in z ; D2 adds bends at -1 and $+1$. Year and seasonal controls are held at sample means. Right: D2 slopes and 95% intervals from eight-release-cycle block resampling. Missing intervals indicate inadequate support.

THE ECONOMIC MECHANISM

Accessible barrels and the ability to receive, store or move oil affect the relative value of earlier delivery. Low aggregate stock can accompany prompt scarcity; high stock can accompany costly carrying arrangements. Leases, flow limits and expectations can weaken either association.

WHAT THE SHAPE CHECK ACTUALLY SAYS

The exploratory joint test does not provide sufficient evidence of a shape difference beyond the linear controlled relationship. D2 versus D1 joint HAC test: $p = 0.7753$; common $n = 481$. The test concerns the historical spread level after year and seasonal controls.

Post-review exploratory extension. The pooled and annual descriptive results had already been inspected. D1/D2 and their diagnostic checks describe this historical sample; they are not an untouched test set, a causal estimate or a return forecast.

03 / WHERE DOES THE RELATIONSHIP FAIL?

The pooled pattern is not a standing rule.

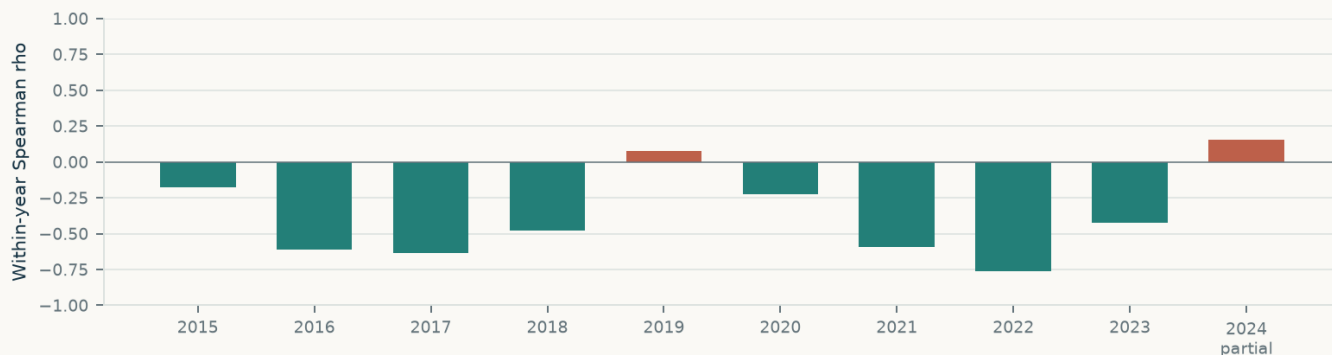


Figure 3. Within-year correlations show that the pooled association varies across periods. 2024 is a partial year: $n = 14$, through 03 Apr 2024.

2019 counterexample: $n = 52$, $\rho = 0.079$. This is a weak association, not evidence of a reliably reversed relationship. The year was selected after reviewing annual results; it is an exploratory diagnostic.

D2 SLOPES: FITTED SPREAD CHANGE PER Z UNIT

Low: -0.124 \$/bbl per z ; 95% block interval $[-0.477, 0.001]$; $n = 133$, years = 7.

Normal: -0.221 \$/bbl per z ; 95% block interval $[-0.609, 0.019]$; $n = 208$, years = 9.

High: -0.040 \$/bbl per z ; 95% block interval $[-0.398, 0.391]$; $n = 140$, years = 5.

Without year intercepts, $p = 2.901e-07$; excluding 2018, $p = 1.326e-06$; excluding 2020, $p = 0.6882$. These diagnostics expose dependence on controls and sample composition. Non-significance does not prove a linear mechanism. All year deletions and block sensitivities are retained.

Evidence boundary: the quote comes before the inventory release. H1 concerns historical market states. H2 needs a fixed-contract future target; H3 separately needs costs and daily cash flows. Neither can be inferred from a pooled correlation or better in-sample fit.

04 / WHAT WOULD A COMMERCIAL ANALYST DO?

Turn a stock reading into a specific check.

2020 / RECEIVING CAPACITY

Released 2020-04-15 for week 2020-04-10: 54.965 million bbl; z 0.39. Quote 2020-04-14: F2-F3 -4.47 \$/bbl. Four-week inventory change: +16.520 million bbl.

Use the inventory report to prioritize an operational capacity check and review delivery exposure. The seasonal threshold alone is insufficient to size a calendar-spread position.

Next check: Ask for tank lease commitments, available injection capacity and incoming pipeline nominations for the relevant delivery month. Nominally empty space is insufficient evidence that a trader can receive barrels.

2023 / REPLENISHMENT EXPECTATIONS

Highest matched spread: +1.67 \$/bbl on quote 2023-10-03. At the inventory trough released 2023-10-18, stocks were 21.013 million bbl and the prior quote was +1.14 \$/bbl. Both extrema are hindsight diagnostics.

Separate a low stock level from an expectation of further spread strengthening. A market-only benchmark is necessary before attributing incremental value to the inventory report.

Next check: Prioritize dated inbound/outbound pipeline nominations, maintenance schedules and nearby physical differentials. They can distinguish a persistent shortage of deliverable barrels from a temporary low stock reading that is expected to rebuild.

WHAT WOULD CHANGE THE VIEW?

2020: verified uncommitted space and injection capacity sufficient for scheduled inflows would weaken a receiving-capacity explanation. 2023: sustained replenishment would weaken persistent scarcity; continued draws and constrained flows would challenge rapid normalization. The current dataset cannot decide between these alternatives.

NEXT DATA, THEN REPRODUCE

Verified month-contract settlements, actual expiries and settlement availability enable the locked B1/B3 forecast comparison. Dated pipeline flows and storage commitments help explain failures. Run validate, build, report using the saved snapshot; hashes verify local consistency, not immutable original publication bytes.

The offline HTML and editable memo retain dated case records, limitations and the full methods. No capital return or real trading performance is claimed.

PRIMARY SOURCE LINKS

2020: original inventory report, released 2020-04-15

2023: original inventory report, released 2023-10-18

EIA petroleum futures prices

NYMEX Light Sweet Crude Oil futures rules